Resources

repmat

Repeat copies of array

Syntax

```
B = repmat(A,n)
B = repmat(A,r1,...,rN)
B = repmat(A,r)
```

Description

`B = repmat(A,n)` returns an array containing `n` copies of `A` in the row and column dimensions. The size of `B` is `size(A)*n` when `A` is a matrix. example

`B = repmat(A,r1,...,rN)` specifies a list of scalars, `r1,..,rN`, that describes how copies of `A` are arranged in each dimension. When `A` has `N` dimensions, the size of `B` is `size(A).*[r1...rN]`. For example, `repmat([1 2; 3 4],2,3)` returns a 4-by-6 matrix. example

`B = repmat(A,r)` specifies the repetition scheme with row vector `r`. For example, `repmat(A,[2 3])` returns the same result as `repmat(A,2,3)`. example

Examples

collapse all

Initialize Matrix with Same Element Value

Create a 3-by-2 matrix whose elements contain the value 10.

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Copy Command

```
A = repmat(10,3,2)
```

Get

```
A = 3×2

    10    10
    10    10
    10    10
```

Square Block Format

Repeat copies of a matrix into a 2-by-2 block arrangement.

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Copy Command

```
A = diag([100 200 300])
```

Get

```
A = 3×3

   100     0     0
     0   200     0
     0     0   300
```

```
B = repmat(A,2)
```

Get

```
B = 6×6

   100     0     0   100     0     0
     0   200     0     0   200     0
     0     0   300     0     0   300
   100     0     0   100     0     0
     0   200     0     0   200     0
     0     0   300     0     0   300
```

Rectangular Block Format

Repeat copies of a matrix into a 2-by-3 block arrangement.

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Copy Command

```
A = diag([100 200 300])
```

Get

```
A = 3×3

   100     0     0
     0   200     0
     0     0   300
```

```
B = repmat(A,2,3)
```

Get

```
B = 6×9

   100     0     0   100     0     0   100     0     0
     0   200     0     0   200     0     0   200     0
     0     0   300     0     0   300     0     0   300
   100     0     0   100     0     0   100     0     0
     0   200     0     0   200     0     0   200     0
     0     0   300     0     0   300     0     0   300
```

3-D Block Array

Repeat copies of a matrix into a 2-by-3-by-2 block arrangement.

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Copy Command

```
A = [1 2; 3 4]
```

Get

```
A = 2×2

     1     2
     3     4
```

```
B = repmat(A,[2 3 2])
```

Get

```
B =
B(:,:,1) =

     1     2     1     2     1     2
     3     4     3     4     3     4
     1     2     1     2     1     2
     3     4     3     4     3     4


B(:,:,2) =

     1     2     1     2     1     2
     3     4     3     4     3     4
     1     2     1     2     1     2
     3     4     3     4     3     4
```

Vertical Stack of Row Vectors

Vertically stack a row vector four times.

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Copy Command

```
A = 1:4;
B = repmat(A,4,1)
```

Get

```
B = 4×4

     1     2     3     4
     1     2     3     4
```

```
     1     2     3     4
     1     2     3     4
```

Horizontal Stack of Column Vectors

Horizontally stack a column vector four times.

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Copy Command

```
A = (1:3)';
B = repmat(A,1,4)
```

Get

```
B = 3×4

     1     1     1     1
     2     2     2     2
     3     3     3     3
```

Tabular Block Format

Create a table with variables `Age` and `Height`.

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Copy Command

```
A = table([39; 26],[70; 63],'VariableNames',{'Age' 'Height'})
```

Get

```
A=2×2 table
    Age    Height
    ___    ______

    39       70
    26       63
```

Repeat copies of the table into a 2-by-3 block format.

```
B = repmat(A,2,3)
```

Get

```
B=4×6 table
    Age    Height    Age_1    Height_1    Age_2    Height_2
    ___    ______    _____    ________    _____    ________

    39       70        39         70        39         70
    26       63        26         63        26         63
```

```
    39    70    39    70    39    70
    26    63    26    63    26    63
```

`repmat` repeats the entries of the table and appends a number to the new variable names.

Combine Vector Elements

Create two column vectors.

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Copy Command

```
A = [1; 3; 5];
B = [2; 4];
```

Get

Generate all element combinations of the two vectors by using `repelem` and `repmat`. Each row of the output `T` is a combination with the first element coming from the first vector and the second element coming from the second vector. This command is equivalent to finding the Cartesian product of two vectors.

```
T = [repelem(A,numel(B)) repmat(B,numel(A),1)]
```

Get

```
T = 6×2

     1     2
     1     4
     3     2
     3     4
     5     2
     5     4
```

Starting in R2023a, you can also use the `combinations` function to generate all element combinations of two vectors.

```
T = combinations(A,B)
```

Get

```
T=6×2 table
    A    B
    _    _

    1    2
    1    4
    3    2
    3    4
    5    2
    5    4
```

Input Arguments

`A` — Input array
scalar | vector | matrix | multidimensional array

Input array, specified as a scalar, vector, matrix, or multidimensional array.

Data Types: `single` | `double` | `int8` | `int16` | `int32` | `int64` | `uint8` | `uint16` | `uint32` | `uint64` | `logical` | `char` | `string` | `struct` | `table` | `datetime` | `duration` | `calendarDuration` | `categorical` | `cell`
Complex Number Support: Yes

`n` — Number of times to repeat input array in row and column dimensions
integer value

Number of times to repeat the input array in the row and column dimensions, specified as an integer value. If `n` is `0` or negative, the result is an empty array.

Data Types: `single` | `double` | `int8` | `int16` | `int32` | `int64` | `uint8` | `uint16` | `uint32` | `uint64`

`r1,...,rN` — Repetition factors for each dimension (as separate arguments)
integer values

Repetition factors for each dimension, specified as separate arguments of integer values. If any repetition factor is `0` or negative, the result is an empty array.

Data Types: `single` | `double` | `int8` | `int16` | `int32` | `int64` | `uint8` | `uint16` | `uint32` | `uint64`

`r` — Vector of repetition factors for each dimension (as a row vector)
integer values

Vector of repetition factors for each dimension, specified as a row vector of integer values. If any value in `r` is `0` or negative, the result is an empty array.

Data Types: `single` | `double` | `int8` | `int16` | `int32` | `int64` | `uint8` | `uint16` | `uint32` | `uint64`

Tips

- To build block arrays by forming the tensor product of the input with an array of ones, use `kron`. For example, to stack the row vector `A = 1:3` four times vertically, you can use `B = kron(A,ones(4,1))`.
- To create block arrays and perform a binary operation in a single pass, use `bsxfun`. In some cases, `bsxfun` provides a simpler and more memory efficient solution. For example, to add the vectors `A = 1:5` and `B = (1:10)'` to produce a 10-by-5 array, use `bsxfun(@plus,A,B)` instead of `repmat(A,10,1) + repmat(B,1,5)`.
- When `A` is a scalar of a certain type, you can use other functions to get the same result as `repmat`.

repmat Syntax	Equivalent Alternative
`repmat(NaN,m,n)`	`NaN(m,n)`
`repmat(single(inf),m,n)`	`inf(m,n,'single')`

repmat Syntax	Equivalent Alternative
`repmat(int8(0),m,n)`	`zeros(m,n,'int8')`
`repmat(uint32(1),m,n)`	`ones(m,n,'uint32')`
`repmat(eps,m,n)`	`eps(ones(m,n))`

Extended Capabilities

expand all

Tall Arrays
Calculate with arrays that have more rows than fit in memory.

C/C++ Code Generation
Generate C and C++ code using MATLAB® Coder™.

GPU Code Generation
Generate CUDA® code for NVIDIA® GPUs using GPU Coder™.

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Thread-Based Environment
Run code in the background using MATLAB® `backgroundPool` or accelerate code with Parallel Computing Toolbox™ `ThreadPool`.

GPU Arrays
Accelerate code by running on a graphics processing unit (GPU) using Parallel Computing Toolbox™.

Distributed Arrays
Partition large arrays across the combined memory of your cluster using Parallel Computing Toolbox™.

Version History

expand all

Introduced before R2006a

R2019b: Some repetition arguments produce error⚠

See Also

`bsxfun` | `kron` | `repelem` | `reshape` | `resize` | `paddata` | `meshgrid` | `ndgrid`

Learn more